

The Reduced-Form Model

From the Quoted Spread to a Measured New York Premium,
and from that Premium to Metal in Motion

Model section, GOLD project

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This section builds the paper's price variable from the number a trader quotes, shows why that number cannot be used as it stands, removes its time dimension by estimating carry rather than assuming it, and arrives at a premium that is defined by a projection of observed prices. It then connects that premium to physical shipments through a threshold. Every object below is a functional of observables. Nothing requires a lease rate, a storage schedule, or a model of tariff risk.

The stance, stated once. The project has two models in it and has been sliding between them. One is *structural*: the spread is written as a function of latent objects — a location premium, a lease rate in each location, a hazard rate for a tariff — and one asks which combinations the data identify. The other is *reduced form*: an object is defined by the regression that produces it, and the paper's claims are claims about that object's behaviour. The two are not interchangeable, and the current headline variable is a hybrid of them, which is the worst of the three options. This section goes fully reduced form and says in §12 what that costs.

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1 The starting point: the quoted spread

Let S_t be the London benchmark price of gold at date t , in US dollars per troy ounce, for *unallocated* metal in a London vault — a claim on a bank’s pooled holding, the gold analogue of a deposit rather than a safe deposit box. Let $F_t(\tau)$ be the price at the same date of a New York futures contract that delivers metal τ days hence, where delivery means a warrant: a title document to specific bars in an exchange-licensed New York vault.

The number the market quotes is the difference:

$$b_t(\tau) = F_t(\tau) - S_t. \quad (1)$$

The economic intuition the paper needs is already here. If New York metal is dear relative to London metal by more than it costs to ship metal between them, someone ships metal, and that shipment is recorded in trade statistics as an export and an import even though nothing about world gold consumption has changed. The paper’s job is to make b into a variable that can carry that argument.

2 Three reasons $b_t(\tau)$ cannot be used as it stands

It is denominated in a unit that is not comparable over time

b is dollars per ounce, and the gold price roughly quadrupled across the sample. The same proportional gap between the two markets is about four times as many dollars at the end of the period as at the start. A regression that pools 2015 dollars with 2026 dollars is pooling incomparable units. This is fixed by working in logs, or equivalently by scaling by S_t .

It contains carry, and carry moves with the delivery calendar

F is a price for delivery later. Even if the two markets are perfectly integrated and metal is worth exactly the same in both places, F exceeds S by the cost of carrying metal from now until delivery, and that cost is proportional to τ .

The difficulty is that τ is not a fixed feature of the data. Gold futures deliver in February, April, June, August, October and December, so the contract the market actually uses has a τ that falls from roughly two months to zero and then jumps back up when traders *roll* — close the expiring contract and open the next one. Carry inherits that sawtooth. At a 4.5% annual holding cost and a \$4,400 gold price, carry is about \$33 an ounce at sixty days and under \$3 at five days. That swing is produced by the delivery calendar and nothing else, and it is the same order of magnitude as the market dislocations the paper is trying to detect.

It mixes a level with a rate

Two economically different things sit inside b .

- A **level**: whatever makes metal in New York worth more than metal in London at a given moment — location, the bar form each market trades, the difference between an exchange warrant and a claim on a bank. A level is paid once. It does not care whether the contract expires in ten days or a hundred.
- A **rate**: financing, storage, and the income forgone by not lending the metal out. A rate accrues per unit of time and therefore scales with τ .

Any single-number summary of b must choose a horizon at which to compare the two, and that choice determines the answer. The project has already seen this: the same day reads as 6.9, 2.8

or 1.8 percentage points annualised depending on whether the curve is read at thirty, ninety or one hundred and eighty days.

The conclusion is not that b is the wrong idea. It is that $b_t(\cdot)$ is a *function* of τ , not a scalar, and the paper's variable has to be a feature of that function rather than one of its values.

3 Carry, and the decision to estimate it

What carry is

Carrying an ounce from today to date τ costs the interest on the money tied up, plus the cost of storing and insuring the metal, less the income the owner forgoes by not lending the metal out. That last term is the *lease rate*: metal, unlike most commodities, can be lent, and the borrower pays for it.

For a macroeconomist the whole structure is covered interest parity with gold in the role of the foreign currency. S is the spot exchange rate, F the forward rate, the dollar interest rate is the domestic rate, and the lease rate is the foreign rate. Carry is the forward points. Deviations from it are the gold analogue of a cross-currency basis.

Why the conventional fix is not reduced form

The usual repair is to convert the spread into the annual financing rate it implies and then subtract an assumed carry:

$$\text{implied rate}_t(\tau) = \left(\frac{F_t(\tau)}{S_t} - 1 \right) \frac{365}{\tau}, \quad (2)$$

$$\text{dislocation}_t(\tau) = \text{implied rate}_t(\tau) - (r_t + \bar{s}). \quad (3)$$

This has two defects, and naming them is the point of this section.

First, it is a hybrid. The implied rate is a reduced-form measurement: it is a transformation of two observed prices. The subtracted term is a structural input: it asserts a storage cost $\bar{s}$ that is not observed and a lease rate of zero that is certainly wrong. So the headline variable is a measurement corrected by assumptions, and it inherits the fragility of the assumptions without gaining any of the discipline that a structural model would provide in exchange. Both errors are also signed and trending: storage is billed in dollars per ounce, so as a *rate* it falls when the gold price rises, and a constant $\bar{s}$ therefore overstates carry by a growing amount across a sample in which gold quadrupled.

Second, it annualises a level. Multiplying by $365/\tau$ is the right operation for something that accrues over time and the wrong one for something paid once. A constant level premium, put through this transformation, produces a fresh sawtooth pointing the other way, and diverges as $\tau \rightarrow 0$. The repair cures one calendar artefact by manufacturing another.

The reduced-form alternative

Do not assume carry. *Estimate* it. On any date the market quotes prices for several delivery months at once, and those prices trace out a term structure: a yield curve for New York gold. The slope of that curve is the market's own carry, revealed by the prices themselves. Nothing has to be assumed about storage, and no interest-rate series is needed to extract it.

4 The estimator

The daily cross-section

On date t , collect every listed contract $i = 1, \dots, n_t$ with a settlement price F_{it} , a horizon τ_{it} and open interest OI_{it} (the number of contracts outstanding, a liquidity measure). Run the weighted least-squares regression

$$\ln F_{it} = a_t + b_t \tau_{it} + u_{it}, \quad w_{it} = \sqrt{OI_{it}}. \quad (4)$$

The functional form follows from compounding, not from a behavioural assumption: if metal is carried at a constant proportional cost c per day then $F(\tau) = F(0)e^{c\tau}$, so $\ln F$ is linear in τ with slope c and intercept $\ln F(0)$. The intercept is the extrapolation of the New York forward curve back to zero horizon — the price today of New York-deliverable metal.

Define the paper’s two price variables:

$$\hat{p}_t \equiv a_t - \ln S_t \quad \text{the New York premium: a level, in log points of spot,} \quad (5)$$

$$\hat{\omega}_t \equiv 365 b_t - r_t \quad \text{excess carry: a rate, in points a year.} \quad (6)$$

What this construction achieves

1. **No rate data enters $\hat{p}_t$.** Carry loads entirely on τ and is absorbed by b_t . Whatever the true financing cost, storage fee or lease rate, they shift the slope, not the intercept. The premium is therefore free of every unobservable that has been blocking the project. Only $\hat{\omega}_t$ needs an interest rate, and only because subtracting r_t is what makes a slope interpretable.
2. **The roll cannot contaminate it.** Nothing here picks a front month. Contracts enter as a cross-section, so the sawtooth has no route in, and the short- τ filters that version 1 needed are no longer load-bearing.
3. **The horizon choice disappears.** We do not read the curve at a chosen maturity; we report its intercept and its slope. The level and the rate are reported as the different objects they are, which removes the arbitrariness documented in §2.
4. **$\hat{p}_t$ is defined by the projection.** This is the crucial point of the whole section. $\hat{p}_t$ is not a parameter of a pricing model that may or may not be true; it is a number computed from prices by a stated rule. It exists, and means the same thing, whether or not any structural account of the gold market is correct. A structural model is then a *claim about what $\hat{p}_t$ should equal* — which is a useful thing to have, and a different thing from the estimand.

Choices that have to be stated, because they are choices

- *Which contracts.* Restrict to $15 \leq \tau \leq 400$ days and open interest above a floor. Contracts inside fifteen days are in the delivery period, where price behaviour is about warrants rather than carry; far-dated and thinly held contracts carry settlement prices that are marked rather than traded. This leaves a median of about six contracts a day.
- *Weights.* Thin contracts have noisier settlements, so weighting by $\sqrt{OI}$ is a proxy for inverse standard deviation. It is an assumption about the error structure, not a result, and the estimates should be shown unweighted as well.
- *Linear, not quadratic.* A horizon-dependent risk — the chance a tariff lands before delivery — would curve the term structure, and fitting τ^2 is tempting. With six points a day a quadratic is fragile and its coefficient is close to collinear with the slope. Baseline linear; curvature as a robustness exercise and as a diagnostic, never as a headline.

- *Daily.* Equation (4) is estimated separately each day rather than pooled with time fixed effects, because both a_t and b_t are the objects of interest and neither is a nuisance.

A sharper measurement of the same estimand

If a forward curve for *London* metal is available at matched dates, the same projection can be run on it: $\ln F_{it}^L = a_t^L + b_t^L \tau_{it} + u_{it}^L$. Then

$$\hat{\pi}_t \equiv a_t - a_t^L \quad (7)$$

is the log premium of New York metal over London metal at zero horizon, measured entirely inside futures markets. It needs no spot benchmark, no interest rate, and no storage figure; and if the two curves settle at the same instant it is free of the timing problem in §6. The slope difference $b_t - b_t^L$ is the cross-location carry differential, which is the natural place to look for a squeeze in the London lending market.

This is not a different model. It is equation (5) measured better, and it should be presented that way: the baseline result uses $\hat{p}_t$, and $\hat{\pi}_t$ is the robustness check that removes three assumptions at once.

5 What the premium contains, and why we do not decompose it

$\hat{p}_t$ contains, at least: the value of metal being in New York rather than London when policy might restrict moving it; the difference between the bar forms the two markets trade; the difference between an exchange warrant and an unallocated claim on a bank; and measurement error from pricing the two legs at different moments.

The reduced-form stance is that we do not attempt to separate these, and the reason is not modesty but identification: with price data alone, a constant institutional wedge and a constant risk premium are observationally equivalent. Attributing a share of $\hat{p}_t$ to any one component would be asserting, not estimating.

What makes this stance workable is that the paper's three claims are invariant to the decomposition:

1. $\hat{p}_t$ moves on dated policy news, which establishes that it is not a constant institutional wedge;
2. $\hat{p}_t$ predicts physical shipments, and only above a threshold, which establishes that it is the price signal that actually moves metal;
3. the shipments it predicts are relocation rather than absorption, which is what makes them a contaminant of trade statistics.

None of the three requires knowing how much of the premium is bar form and how much is tariff risk. What we give up is equally clear and should be written in the paper rather than left for a referee to notice: we cannot decompose the premium, so we make no statement about the price of tariff risk as such, and no welfare claim that would require one.

6 Measurement error, and the one structural fact we do use

The two legs are not observed at the same instant. The London benchmark is struck in the afternoon London auction; the New York settlement is calculated about three and a half hours later. Let ε_t be gold's return over that window.

One fact about the trading mechanism does real work here, and it is worth keeping precisely because it is a fact about the mechanism rather than an assumption about behaviour: *every contract settles at the same instant*. So ε_t multiplies all of that day's futures prices equally. In logs,

$$\widehat{a}_t = a_t + \varepsilon_t, \quad \widehat{b}_t = b_t. \quad (8)$$

The timing error is a pure *level* contaminant. It biases the premium by $\mathbb{E}[\varepsilon_t]$ and adds $\text{Var}(\varepsilon_t)$ to its variance, and it leaves the estimated carry slope untouched.

Two consequences follow, and they determine the frequency at which each part of the paper should be run.

- **As an outcome**, in the news regression, classical measurement error inflates standard errors but does not bias the coefficient — provided the news does not arrive inside the three-and-a-half-hour window, which for some announcements it does. The defence is an event window that opens before and closes after the affected settlement, plus re-timing the New York leg to the London auction instant where intraday data allow.
- **As a regressor**, in the flow equation, the same error attenuates the slope towards zero and, because the specification has a kink, smooths the kink and biases the estimated threshold. The defences are averaging to monthly frequency, which shrinks the noise variance by the number of days in the month; re-timing; and instrumenting the premium with dated announcements.

That the news regression belongs at daily frequency and the flow regression at monthly is therefore not a convenience. It falls out of where the measurement error lands.

7 From premium to metal: the second reduced form

Write the premium in dollars per ounce, $P_t = \widehat{p}_t S_t$, and let Q_t be tonnes shipped along the relevant corridor in month t . Metal moves only when the premium covers the all-in cost of moving it: freight, insurance, recasting into the bar form the destination market accepts, and the financing of metal in transit. Below that cost, nothing happens; above it, the trade is profitable and scales with how far above it sits. The reduced-form representation is a hinge:

$$Q_t = \beta^W \max(0, P_t - \kappa^W) - \beta^E \max(0, -P_t - \kappa^E) + \gamma' x_t + u_t. \quad (9)$$

Four features of this deserve a sentence each.

Dollars, not rates. Freight, insurance and refining are charged by weight, so the threshold is naturally a dollar amount per ounce and should not be expressed as a percentage. This is a prior about the technology, not a fitted result, and the paper should say so plainly; the two specifications are only distinguishable when the gold price moves a lot.

A kink, not a slope. A linear specification lets the many months in which nothing moves pull down the estimated response of the months in which a great deal moves. The kink is also the economically interesting parameter: κ is the transatlantic transfer cost, which is not published anywhere, so estimating it is a contribution in its own right rather than a nuisance.

Two sides, separately. Above the band New York is dear and metal flows west; below it New York is cheap and metal flows east. Both regimes occur. A single dummy or a symmetric specification would impose that the two thresholds and the two slopes are equal, which is exactly the kind of restriction the data can test.

Daily hinge, then aggregate. Because $\max(0, \cdot)$ is convex, applying the hinge to a monthly average premium is not the same as averaging the daily hinge, and it systematically understates

the response: a month containing a few days far above the threshold and many days below it looks, in its average, like a month that never crossed. Apply the hinge day by day and aggregate afterwards.

Two further shifters belong in x_t , and both are reduced-form controls rather than structural amendments. Capacity — refineries and secure freight can only move so much per month — censors Q from above, so a period of sustained high premia with flat flows is not evidence against the mechanism. And the stock of metal already inside the destination market matters: an arbitrage can be closed by re-warranting metal that is already in New York, which produces no cross-border shipment at all. That is not a defect of the specification but the central asymmetry of the whole project, and it belongs in the flow equation as a control and in the accounting as a bound.

8 How the pieces fit together

Step	What is estimated	Frequency	What it establishes
1	Cross-sectional projection (4); the premium (5) and excess carry (6)	daily	a price variable free of carry, the roll and any horizon choice
2	Response of $\Delta\hat{p}_t$ to signed, timed policy announcements	daily	the premium is priced policy risk, not a constant wedge
3	The hinge (9): β and κ	monthly	the premium moves metal, and only once it clears a cost
4	Fitted flows against absorption benchmarks and vault stocks	monthly	those flows are relocation, not consumption
5	Relocation flows applied to reported trade aggregates	monthly	the size of the contamination — the paper's payoff

Steps 2 and 3 can be run jointly, with the announcements instrumenting the premium in the flow equation. That buys consistency if the premium is measured with error, at the cost of an exclusion restriction: announcements must affect shipments only through the premium. Three things threaten it, and each has a reduced-form remedy rather than a structural one — safe-haven demand shifting flows directly, which is why ETF and central-bank series enter as controls; precautionary relocation by owners who would move metal whatever the price, which is untestable and must be conceded; and country-wide tariff measures that affect all trade with a partner, which is why gold-specific events are preferred to broad ones.

9 Arithmetic illustration of the definitions

No estimation is reported in this note. To make the mapping concrete, here is one date run through the definitions, using quantities computed earlier in the project.

On 29 January 2025 the fitted curve had an intercept implying a zero-horizon New York price of \$2,769.38 against a London benchmark of \$2,756.30. Then

$$\hat{p}_t = \ln\left(\frac{2769.38}{2756.30}\right) = 0.473\%, \quad P_t = \hat{p}_t S_t = \$13.08 \text{ per ounce},$$

which, against a transfer cost of order \$3 an ounce, sits well inside the region where equation (9) predicts metal moves. The fitted slope on the same day was 5.22% a year, so with an overnight rate of 4.35% the excess carry was $\hat{\omega}_t \approx +0.9$ points before any storage adjustment.

Two scale conversions make these numbers readable. One per cent of spot at a \$4,500 gold price is \$45 an ounce, or \$1.45 million a tonne. A \$3 per ounce threshold is about \$96,000 a tonne, which is where air freight, insurance and recasting plausibly sit — a sanity check on the estimand, not a validation of it.

10 Notation, units and signs

Symbol	Meaning	Unit
S_t	London benchmark, unallocated metal	\$/oz
$F_t(\tau)$	New York futures price, delivery in τ days	\$/oz
τ	calendar days to the delivery-relevant date	days
a_t, b_t	intercept and slope of the daily curve (4)	log points; log points/day
$\hat{p}_t$	New York premium, $a_t - \ln S_t$	log points of spot
P_t	the same premium in money, $\hat{p}_t S_t$	\$/oz
$\hat{\omega}_t$	excess carry, $365b_t - r_t$	points a year
κ	transfer cost threshold	\$/oz
β	flow response above the threshold	tonnes per month per \$/oz
Q_t	metal shipped	tonnes a month

Conventions. Rates are annualised decimals and differences between rates are quoted in percentage points. Day count is 365 throughout; if an overnight rate quoted on a 360-day basis is used in (6) it must be scaled by 365/360 first. Mass is tonnes, with 1 t = 32,150.7 oz, and prices are converted at the boundary rather than inside the analysis.

Signs. $\hat{p}_t > 0$ means New York is dear, so metal flows west, towards New York. $\hat{p}_t < 0$ means the reverse. Both regimes occur in the sample, and (9) is written to handle both rather than assuming the westward case.

11 Assumptions, and where each one fails

Assumption	Where it enters	How it fails
The log forward curve is linear in the horizon	(4)	horizon-dependent risk curves it; a tariff hazard bends it downward, loading onto the slope and biasing $\hat{\omega}_t$, not $\hat{p}_t$
The timing shock is common to all contracts	§6	fails if contracts settle at different instants, or if a far month is marked rather than traded; then the error tilts the curve and contaminates $\hat{p}_t$ as well
Settlement noise falls with open interest	weights in (4)	fails when a thin far contract is pinned to a model price and so looks <i>less</i> noisy than it is
The threshold is constant in dollars within the estimation window	(9)	freight and insurance move with fuel and with wartime risk premia; a level shift in κ mimics a change in β
Shipments are recorded in the month the premium was earned	aggregation	exporters record on departure and importers on arrival, so month-boundary crossings misalign the two sides
Announcements move flows only through the premium	step 3 of §8	safe-haven demand and precautionary relocation both violate it; the second is untestable with available data
The premium is a price signal, not a quantity constraint	(9)	when capacity binds, Q stops responding and β is attenuated; treat the ceiling explicitly

12 What we give up by not going structural

A structural version of this exercise would write the premium as a function of a location risk premium, a lease rate in each market and a hazard rate for tariff imposition, and would estimate those objects. It is worth being explicit about why the paper does not do that.

1. **The central parameter has no data behind it.** There is no reported New York lending market for gold at any price. In the structural system the New York lease rate appears only in combination with the storage cost, so the futures curve alone can never separate them. A structural estimate would be an assumption wearing an estimate's clothes.
2. **The components of the premium are observationally equivalent in price data.** Location risk, bar-form scarcity and warrant-versus-claim trust all enter as a level. No amount of price data separates them; only owner-level or transaction-level data would, and it does not exist publicly.
3. **The claims the paper actually makes do not need the decomposition.** Everything in §5 survives any allocation of the premium across its components.

So the structural system belongs in an appendix, where it does two honest jobs: it explains what $\hat{p}_t$ and $\hat{\omega}_t$ would equal under an explicit pricing model, which is how the reader knows

what sign to expect; and it states precisely which combinations are not identified, which is how the reader knows the main text is not quietly claiming them.

Three consequences for the paper’s prose, which are the practical content of taking this stance.

- *Retire “dislocation” as the headline variable.* It is the hybrid object of §3: a measurement minus an assumption. Report $\hat{p}_t$ and $\hat{\omega}_t$ instead, which are both projections of observables. If a single annualised number is wanted for comparability with market commentary, report it in a footnote with the assumed storage rate stated next to it.
- *Never attribute a share of $\hat{p}_t$ to a named component.* Sentences of the form “the tariff premium reached x per cent” should read “the New York premium reached x per cent”, with the tariff attribution carried by the event study rather than by the label.
- *Report the level and the rate separately, always.* They answer different questions — how much more a dealer will pay to have metal in New York today, versus how expensive it is to hold it there — and collapsing them into one number was the source of the confusion this section exists to remove.